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Excess risk of chronic health conditions in Black adolescent and young adult cancer survivors

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Abstract

Background—The US population of adolescent and young adult (age 15–39 years at diagnosis) cancer survivors is growing. Previous studies have identified racial and ethnic disparities in survival and health outcomes in racially minoritized survivors, including Black survivors, compared with White survivors. However, comparisons should be made between those of the same race or ethnicity with and without a history of AYA cancer to fully understand the association of a cancer diagnosis with socioeconomic status (SES) and health outcomes within a minoritized population.

Methods—Non-Hispanic Black AYA cancer survivors and non-Hispanic Black age- and sex-matched controls were identified from self-reported data from the National Health Interview Survey (2009–2018). SES factors and chronic health conditions prevalence were compared between survivors and controls using chi-square tests. Survey-weighted logistic regression models were used to determine odds of chronic conditions by SES factors within and between survivors and controls. Interactions between each variable and cancer group were assessed.

Results—A total of 445 survivors and 4450 controls were included. Survivors were less likely than controls to be married, have family income >45K/year, have completed a bachelor's degree or higher, and have private insurance. Survivors had higher odds than controls of having at least one (odds ratio (OR): 7.02, $p<0.001$) and ≥3 (OR: 4.44, $p<0.001$) chronic conditions. Survivors had higher odds of each chronic condition assessed including cardiovascular disease, diabetes, and hypertension. Survivors had higher odds of having chronic health conditions compared with controls across all SES variables.

Conclusions—A cancer diagnosis during adolescence and young adulthood is associated with poor SES outcomes and increased odds of comorbidities within the Black population, thus further exacerbating existing disparities.

Implications for Cancer Survivors—Black AYA cancer survivors have a very high risk of developing chronic health conditions after cancer treatment and interventions are needed to improve long-term health outcomes for this population.

Keywords

Adolescent and young adult; AYA; Cancer survivors; Disparities; Chronic conditions

Introduction

The incidence of cancer among adolescents and young adults (AYAs, aged 15–39 years at diagnosis) in the USA is steadily rising with approximately 90,000 new diagnoses per year [1]. Detection and treatment improvements have led to increased survival, with 5-year survival rates now >80% [2, 3]. As AYA cancer survivors are living decades past diagnosis, they are at risk for both treatment and age related comorbidities, which translates into an increased chronic disease burden compared with the general population [4].

AYA cancer survivors have about a 50% increased incidence of any comorbidity compared with the general population and common chronic conditions in AYA survivors including hypertension, dyslipidemia, cardiovascular disease (CVD), and diabetes [4]. Compared

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with non-Hispanic White survivors, racially minoritized survivors of AYA cancer are at increased risk for many common chronic conditions [4–7]. In the general population, there are also disparities in the incidence of chronic conditions by race and ethnicity. For example, compared with White individuals in the general population, Black individuals have increased likelihood of diabetes, hypertension, and CVD [8, 9]. Thus, it is difficult to discern whether disparities seen in survivors are simply reflective of those in the general population or if an AYA cancer diagnosis further exacerbates these existing disparities. Additionally, while not well studied in the AYA population, previous studies among childhood cancer survivors have found racial and ethnic disparities in markers of socioeconomic status (SES), with Black survivors reporting lower levels of income, education attainment, and insurance coverage compared with White survivors [10]. Racial and ethnic disparities in SES are also present in the general population; thus, comparison of SES between AYA survivors and those without a history of cancer within the same racial or ethnic group is similarly needed.

To address this gap, we evaluated the prevalence of chronic conditions among Black AYA cancer survivors compared with Black individuals without a cancer history. We also compared SES markers including income, insurance status, education attainment, and marital status between Black survivors and controls and evaluated the association of sociodemographic factors on risk of chronic conditions within each group.

Methods

Data source

In order to investigate the association of SES factors on the development of chronic health conditions in non-Hispanic Black or African American (hereafter referred to as “Black”) AYA cancer survivors, data from the 2009–2018 National Health Interview Survey (NHIS) were analyzed. The NHIS is a cross-sectional household survey and principal source of information regarding the health of the non-institutionalized civilian population in the USA [11]. The data used in this study were from the NHIS Sample Adult File and NHIS Public-Use Person File.

Sample

Out of a total sample of 956,922 participants in the 2009–2018 NHIS survey, we selected individuals who self-identified as Black AYA cancer survivors using the following questions: “Do you consider yourself to be Black/African American?,” “Have you ever been informed by a physician or other health practitioner that you had cancer?,” and “How old were you when you were first diagnosed with cancer?” Based on the responses to those questions, we defined AYA cancer survivors as individuals who participated in the survey and were 18 years of age or older, having received any cancer diagnosis between 15 and 39 years of age. As a reference group, Black individuals without any prior history of cancer were selected as controls, with a ratio of 10:1 to Black AYA cancer survivors, based on age at survey completion (continuous), sex, and year of survey. This study included a final sample size of 445 Black AYA survivors and 4450 matched controls. For the multivariable analyses, 7 survivors were excluded due to missing insurance and education data, and 18

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matched controls were excluded due to missing data on chronic health conditions, including 2 controls with missing insurance data.

Variables and outcomes

The existence of chronic health conditions was measured by asking "Have you ever been told by a doctor or other health professional that you had any of the following: Chronic health conditions other than cancer include stroke, CVD, kidney disease, hypertension, arthritis, liver disease, pulmonary disease, and diabetes?" The following variables were incorporated as covariates: age at survey completion, dummy coded as 18–39 years (reference), 40–64 years, and 65–85 years; sex (0=male, 1=female); marital status, dummy coded as married or living with a partner (reference), divorced/separated/widowed, and never married; education, coded as less than bachelor's degree (reference), and bachelor's degree or higher; family income, dummy coded as $45K or higher (reference), less than $25K, and $25K–$44,999 based on the distribution of income in the USA [12]; insurance, coded as private (reference), public, and uninsured.

Statistical analysis

Frequencies and weighted percentage of the study variables were estimated. Chi-square and *t*-tests were used to compare the study variables between Black survivors and Black control groups. To examine the association between chronic health conditions and covariates, we conducted survey-weighted logistic regression to model the log-odds of chronic health conditions. In these regression models, the dependent variable was the existence of chronic health condition (0=no, 1=yes), and independent variables included sociodemographic factors such as age at survey completion, sex, marital status, educational attainment, family income, and health insurance, together with the interactions between each variable and cancer group (i.e., survivors versus controls). To evaluate the differences between reference levels and discrete variable levels within each cancer group, contrasts were used, and Dunnett-adjusted *p*-values were used to account for multiple comparisons. Furthermore, the study assessed the differences in discrete variable levels (interaction effects) between the cancer groups by using contrasts, and Hommel method was used to adjust the *p*-values for multiple comparisons. We provided the odds ratio (OR) along with confidence intervals of 95%. NHIS sampling procedures were accounted for by using sample person weights, primary sampling unit, and strata, and weights from 2009 to 2018 were divided by 10 [13]. Two-sided statistical significance was evaluated at the alpha level of 0.05. All statistical analyses utilized R statistical software (version 4.2.1).

Results

Sociodemographic characteristics

Table 1 presents sociodemographic descriptive statistics of 445 Black AYA cancer survivors and 4450 matched controls. In female survivors, the most prevalent cancer was cervical cancer (n=118, 32%), followed by breast (n=81, 22%) and uterine cancer (n=63, 17%). Among male survivors, lymphoma was the most prevalent cancer type (n=22, 28%), followed by prostate (n=9, 11%) and lung cancer (n=8, 10%, Supplemental Table 1). The mean age of cancer survivors at diagnosis was 29 years (range 18–39). In both survivors

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and controls, 81% were female. Compared to controls, survivors were more likely to be divorced/separated/widowed, attain lower education levels have lower annual family income, and have public insurance. The number of family members in the survivor group was less than the control group (2.22 vs 2.85, $p<.001$) and the average family income of married individuals in both groups was greater than that of unmarried individuals.

Risk of chronic health conditions in AYA cancer survivors

Survivors had significantly higher likelihood than controls of having at least one chronic health condition (81% vs. 32%, OR: 7.28, adjusted OR [aOR]: 7.02, $p<.001$) and of having three or more chronic health conditions (31% vs. 9%, aOR: 4.44 $p<.001$) (Table 2). In addition, survivors had greater likelihood than controls of reporting every chronic health condition, including CVD (aOR: 4.32, $p<.001$), kidney disease (aOR: 3.66, $p<.001$), hypertension (aOR: 4.19, $p<.001$), arthritis (aOR: 3.94, $p<.001$), liver disease (aOR: 3.92, $p<.001$), pulmonary disease (aOR: 2.99, $p<.001$), stroke (aOR: 2.88, $p<.001$), and diabetes (aOR: 2.29, $p<.001$).

Chronic health conditions by age and sex

Older age at time of survey was associated with increased odds of chronic health conditions in both survivors and controls (Table 3), and, compared with controls, AYA survivors had higher odds of chronic conditions within each age group assessed (18–39 years, 40–64 years, 65–85 years, Table 4, Fig. 1A). Interaction contrasts showed that age impacts odds of chronic conditions to a lesser extent in AYA survivors than controls ($p=0.019$, Supplemental Table 2).

Sex was not significantly associated with the existence of chronic health conditions in either controls or survivors (Table 3). However, both male and female survivors had significantly higher odds of having chronic health conditions compared with male and female controls, respectively (male: OR=14.57, $p<.001$; female: OR=8.30, $p<.001$) (Table 4, Fig. 1B). No interaction was observed by sex at birth between survivors and controls ($p=0.190$, Supplemental Table 2), so there was no significant evidence that the difference in males differed from the difference in females.

Chronic health conditions by marital status and education

Marital status was significantly associated with the existence of chronic health conditions in controls, but this association was not seen among AYA survivors (Table 3). There were significant differences in the odds of chronic health conditions based on marital status between survivors and controls (Table 4, Fig. 1C). AYA survivors who were married or living with a partner, divorced/separated/widowed, and never married all had higher odds of having a chronic health condition compared with controls of the same marital status. A test for interaction was not statistically significant ($p=0.150$ and 0.250, Supplemental Table 2). Educational attainment was not significantly associated with existence of chronic health conditions in either survivors or controls (Table 3). There were significant variations in the odds of having a chronic health condition based on the level of education when comparing survivors and controls (Table 4, Fig. 1D). The test for interaction by the level of education

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between survivors and controls was not statistically significant (p=0.080, Supplemental Table 2).

Chronic health conditions by income and insurance status

Family income was significantly associated with the presence of chronic health conditions in controls, but not in survivors (Table 3). There were significant differences in the odds of chronic health condition by family income level when comparing survivors and controls (Table 4, Fig. 1E). AYA survivors who had an annual family income of less than $25K, between $25K and $45k, and more than $45K had significantly greater odds of having chronic health conditions compared with controls in the same family income brackets. The test for interaction by annual family income level between survivors and controls was not statistically significant (p=0.140 and 0.100, Supplemental Table 2). While insurance status was associated with odds of chronic conditions in controls, this was not seen among survivors (Table 3). However, Black AYA survivors had higher odds of chronic conditions than Black controls within each insurance status bracket (Table 4, Fig. 1F). The test for interaction did not reach statistical significance (p=0.280, Supplemental Table 2).

Discussion

In the current study, we found that Black AYA cancer survivors differ from age- and sex-matched Black individuals without a history of cancer in both markers of SES and diagnosis of chronic health conditions. Specifically, Black survivors were less likely to be married, and had lower family income and education attainment, greater proportion of public insurance coverage, and higher prevalence of chronic conditions compared with Black controls. Importantly, this study compares Black AYA cancer survivors to Black individuals without a history of cancer, allowing for a better evaluation of the impact of an AYA cancer diagnosis on SES and comorbidity outcomes within the Black population.

Socioeconomic disparities

The SES disparities seen in the current study in Black AYA cancer survivors compared with Black controls highlight that an AYA cancer diagnosis may further exacerbate SES disparities in an already-vulnerable population. In the general population, Black individuals are more likely to have lower household income, less likely to be married, more likely to have lower education attainment, and less likely to be insured than White individuals. [14–17]

A cancer diagnosis at any age can have lasting impacts on SES; however, due to developmental and life stage, a diagnosis as an AYA carries particularly high risk of adverse SES impacts. AYAs are more likely to be under or uninsured than other age groups, and, due to earlier career phase than older patients, may experience greater productivity loss with impacts on both short-term finances as well as long-term earnings potential [18]. At baseline in the USA, Black individuals earn less than White individuals [15]. Our findings demonstrate that cancer as an AYA is associated with lower household income in Black survivors compared with Black individuals without a history of cancer, thus widening existing financial disparities. Family income is associated with the size of the household

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and marital status; our findings show that survivors were more likely than controls to be unmarried and to report low income. The financial impacts of cancer as an AYA go beyond reduced household income, as survivors experiencing financial toxicity are more likely to forego recommended medical follow-up and screenings as well as delay or avoid filling prescribed medications [18]. This points to the importance of early screening for financial distress as well as the development of multidisciplinary culturally adapted financial navigation protocols, particularly among minoritized AYA patients and survivors [19].

Finally, while there was no evidence of differences in uninsured coverage rate, we found that Black survivors were more likely than Black individuals without a history of cancer to be publically insured. Survivors of AYA cancer may be more likely than those without a cancer history to meet public insurance eligibility requirements. Given that the timeframe of the survey responses included in the current study encompasses the years immediately surrounding enactment of the Affordable Care Act, it may also be possible that AYA survivors had increased interaction with the medical system and thus may have received more education and assistance around public insurance enrollment than individuals without a history of cancer.

Chronic health conditions

Previous studies assessing comorbidities in AYA cancer survivors have found an increased burden of chronic conditions in survivors compared with controls, and, within AYA survivors, have found that Black survivors have increased risk of certain chronic conditions compared with White survivors [4, 20–22]. In the current study, we found that Black AYA cancer survivors were more likely to have a chronic condition diagnosis than Black individuals without a history of cancer, and had significantly higher prevalence of each condition assessed, including diabetes, CVD, pulmonary disease, hypertension, arthritis, liver disease, stroke, and kidney disease. Similarly, in the overall AYA survivor population, recent data have shown incidence rates per 1000 person years at risk of 10.4, 4.2, 5.5, 16.1, and 5.2 for diabetes, CVD, pulmonary disease, hypertension, and liver disease, respectively, all significantly higher than incidence rates in controls [4]. In the US general population, data have shown that Black individuals have a higher prevalence than White individuals of many chronic conditions including hypertension, diabetes, and CVD [23, 24]. In contrast, Black individuals in the general population have a lower prevalence of non-alcoholic chronic liver disease and a similar prevalence of early stage kidney disease, though a higher prevalence of end-stage kidney disease compared with White individuals in the general population [25, 26].

While there are racial differences in mortality risk from many different chronic conditions, for Black individuals, differential mortality from hypertension and diabetes account for a large proportion of disparities in overall mortality compared with White individuals in the general population [27]. Thus, the increased prevalence of these diagnoses found in Black AYA survivors compared with Black controls places an already-vulnerable population at heightened risk of mortality in addition to morbidity. Black cancer survivors are less likely than White cancer survivors to report medical specialist care, less likely to receive recommended diabetes management care, and more likely to forgo prescription medications

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because of cost [28–30], all placing this group at further risk of adverse outcomes. Higher number of chronic conditions in cancer survivors is also associated with increased mortality rate [31], and we found that Black survivors had about 4-fold higher odds of having multiple chronic condition diagnoses compared with Black controls. Additionally, compared with controls, older age at the time of survey completion was disproportionately associated with increased odds of chronic health conditions compared with younger age among Black AYA survivors compared with controls. Premature aging related to cancer therapies can lead to earlier accumulation of age-related comorbidities with higher risk of mortality compared with the general population [32], and our study suggests similar findings in the Black AYA survivor population. This points to the need for early screening and management, particularly in minoritized populations.

Association of socioeconomic factors with chronic health conditions

In multiple logistic regression models, we found that SES factors were not significantly associated with increased odds of chronic conditions in Black AYA cancer survivors. However, in Black controls, insurance status, marital status, and household income were all associated with increased odds of chronic conditions. Previous studies among AYA cancer survivors have found that those with public insurance were at higher risk for chronic conditions and have also found increased chronic conditions risk among survivors with lower SES [6, 7, 33]. It is unclear why markers of SES were not associated with differential odds of chronic conditions in Black AYA survivors in the current study, though it is possible that our sample size of survivors limited detection of these differences. Because chronic health conditions are so prevalent in the Black AYA survivor population, very large sample sizes are needed to see small differences. Further studies are needed to determine whether this finding persists across different cohorts. Importantly, our findings do show that Black AYA survivors across all SES groups are at increased risk of chronic conditions, highlighting that this a vulnerable group needing close surveillance and improved access to high quality survivorship care.

Study limitations

There are several strengths and limitations to the current study. The NHIS consists of self-reported data, and thus there is potential for both misreporting of medical conditions and underreporting as undiagnosed conditions may be missed. It is also possible that AYA survivors are more engaged with the healthcare system and thus more likely to know and report their chronic conditions. However, the NHIS remains the primary recommended source for monitoring the US population health status including disease prevalence as set forth in the Healthy People 2030 framework [34]. As the NHIS does not capture cancer treatment data, this could not be included in the current analysis though it is known to differentially impact chronic conditions risk. Additionally, the NHIS data does not contain the timing of onset of chronic health conditions. Thus, the time from cancer diagnosis to chronic health conditions could not be analyzed in this study. Finally, our study had a predominance of females which may have impacted our ability to detect differences in odds of chronic conditions by sex. Prior studies among AYA cancer survivors utilizing US population-based self-report datasets including NHIS have also been comprised of a majority female population [20, 35, 36]. This is likely partly reflective of the sex-based

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differences in cancer incidence in AYAs. Among Black AYAs specifically, females have nearly double the rate of new cancer diagnoses compare with males [37]. The comparison of a Black AYA survivor population to age- and sex-matched controls of the same race is a strength of the current study as it allows for a better understanding of the impact of an AYA cancer diagnosis on a minoritized population.

Conclusion

In summary, we found that Black AYA cancer survivors face SES disparities including lower rates of marriage, lower education attainment, lower household income, and higher proportion of public health insurance compared with Black controls. Additionally, Black AYA survivors had significantly higher odds of having one or more chronic health conditions. Importantly, we found these disparities in an already-minoritized population, suggesting that an AYA cancer diagnosis can further exacerbate disparities in Black individuals. Comprehensive follow-up care and screening for late effects are necessary for early detection and treatment of chronic conditions; however, barriers to care access have been identified in both Black survivors and those that are underinsured [38]. In breast cancer survivors, culturally targeted interventions have shown success in increasing knowledge of survivorship care as well as improving health behaviors in Black survivors [39–41]. Similarly targeted interventions among Black AYA cancer survivors as well as structural changes aimed at increasing access to survivorship care among minoritized survivors are urgently needed.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Data Availability

All data are publically available in the NHIS database.

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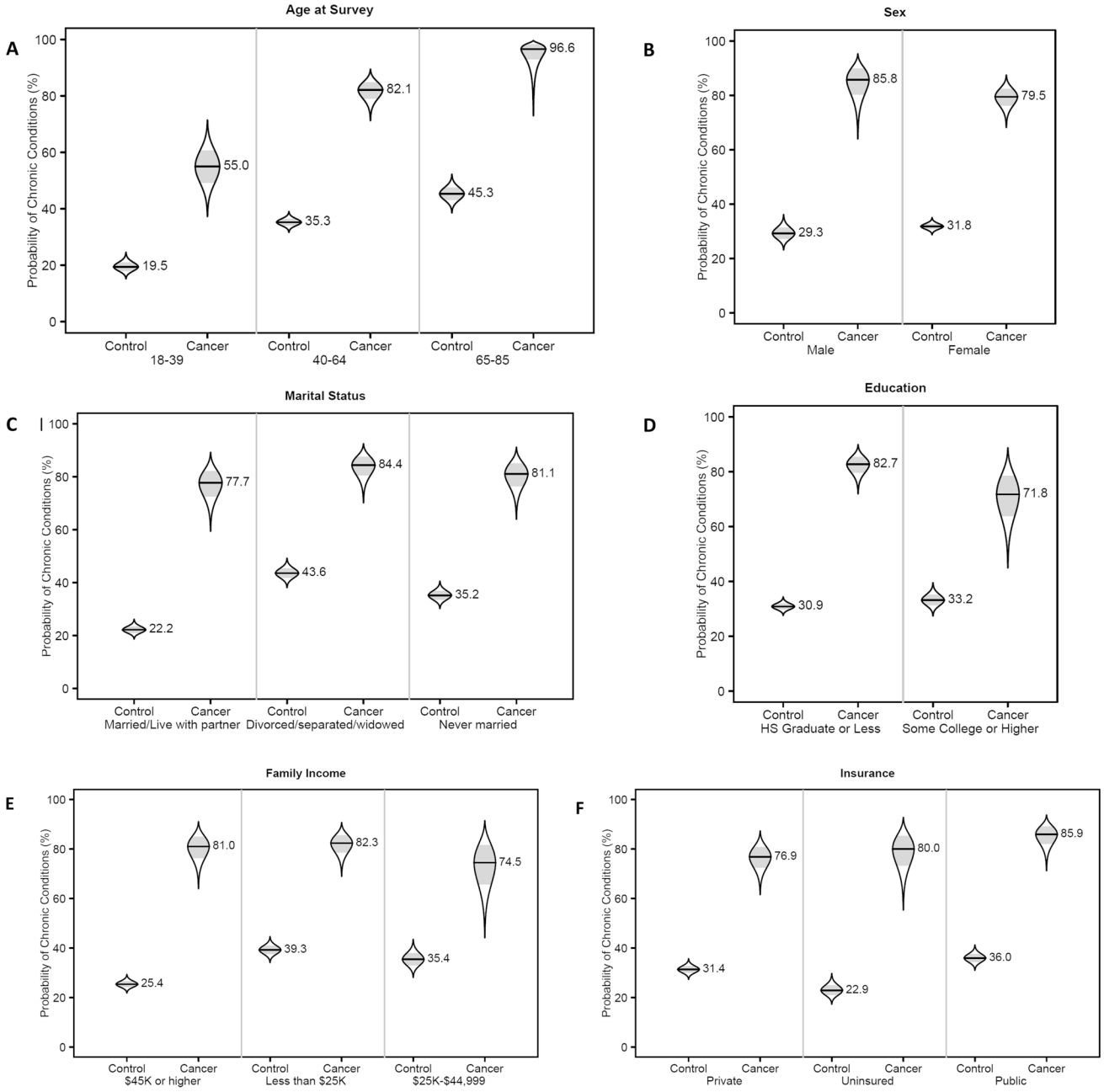

Fig. 1.

Probability of having a chronic health condition with relation to: age at survey (**A**), sex (**B**), marital status (**C**), education (**D**), family income (**E**), and insurance (**F**), by cancer group. Cats-eye plots demonstrate the normal distributions of the model-adjusted means with shaded ± standard error intervals; the width of each cats-eye plot is arbitrary, and the normal distributions are simply the classic, normal bell-curve turned sideways and mirrored horizontally. Model-adjusted means are weighted proportionally to covariate marginal frequencies and are transformed to the probability scale

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Table 1

Sociodemographic characteristics of Black AYA cancer survivors and their matched controls

		Black controls (*N*=4450)*			Black AYA cancer survivors (*N*=445)			*p*
		Mean (SD)	***n***	**Wt%**	**Mean (SD)**	***n***	**Wt%**	
Age at survey		48.74 (14.86)			48.75 (14.89)			0.998
	18–39 years		1360	34.0		136	32.3	0.678
	40–64 years		2320	51.3		232	51.4	
	65–85 years		770	14.7		77	16.3	
Age at diagnosis					28.93 (6.7)			
	15–19 years					44	11.1	
	20–29 years					170	38.1	
	30–39 years					231	50.7	
Sex	Female		3650	80.9		365	80.5	0.826
	Male		800	19.1		80	19.5	
Marital status	Married or living with partner		1958	44.4		132	29.2	**<.001**
	Divorced/separated/widowed		1234	26.0		169	37.8	
	Never married		1258	29.7		144	32.9	
Education	Less than High school		744	15.6		75	16.3	**0.019**
	High school graduate or GED		1258	27.4		106	24.4	
	Some college		1512	34.6		187	41.2	
	Bachelor’s degree or higher		936	22.4		76	18.1	
	Less than bachelor’s degree		3514	77.6		368	81.9	**0.042**
	Bachelor’s degree or higher		936	22.4		76	18.1	
Family income		49,614 (43,832)			37,663 (37,477)			**<.001**
	Less than $25,000		1550	33.6		226	49.7	**<.001**
	$25,000–$44,999		1161	26.1		103	21.9	
	$45,000 or higher		1739	40.2		116	28.4	
Insurance	Uninsured		725	16.5		71	15.1	0.454
	Insured		3723	83.5		373	84.9	
	Uninsured		725	16.5		71	15.1	**<.001**
	Public		1441	30.5		191	43.0	

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	Black controls (*N*=4450)*			Black AYA cancer survivors (*N*=445)			*p*
	Mean (SD)	***n***	**Wt%**	**Mean (SD)**	***n***	**Wt%**	
Private		2282	53.0		182	41.8	

p values < 0.05 are noted in bold

AYA adolescent and young adult, *SD* standard deviation, *Wt%* weighted percentage

*Two controls were missing insurance data and 7 survivors were missing insurance and/or education data

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Table 2

Prevalence and adjusted odds ratios of chronic health conditions in Black AYA cancer survivors

Number of diagnosed chronic condition	Controls^ (*N*=4432)		AYA cancer survivors (*N*=445)		Unadjusted			Adjusted		
	n	Wt%	*n*	Wt%	OR	CI	*p*	OR*	CI	*p*
0	2973	68.0	99	23.2			**<.001**			
1	599	13.9	105	24.1						
2	421	9.3	97	21.8						
3+	439	8.9	144	30.9						
0	2973	68.0	99	23.2	7.28	5.56–9.54	**<.001**	7.02	5.48–8.99	**<.001**
1+	1459	32.0	346	80.8						
0–2	3993	91.1	301	69.1	4.60	3.57–5.91	**<.001**	4.44	3.26–6.05	**<.001**
3+	439	8.9	144	30.9					5.48–8.99	
Types of diagnosed chronic conditions										
Cardiovascular disease	271	5.5	113	23.4	5.23	4.10–6.71	**<.001**	4.32	3.19–5.85	**<.001**
Kidney disease	62	1.2	26	5.7	5.10	2.75–7.02	**<.001**	3.66	2.19–6.12	**<.001**
Hypertension	1049	22.5	258	56.4	4.65	3.66–5.47	**<.001**	4.19	3.25–5.39	**<.001**
Arthritis	682	14.5	194	42.4	4.37	3.48–5.24	**<.001**	3.94	3.03–5.14	**<.001**
Liver disease	65	1.5	28	6.5	4.70	2.88–7.14	**<.001**	3.92	2.07–7.42	**<.001**
Pulmonary disease	356	8.2	104	23.9	3.51	2.75–4.48	**<.001**	2.99	2.19–4.09	**<.001**
Stroke	111	2.5	43	8.4	3.64	2.90–6.03	**<.001**	2.88	1.77–4.70	**<.001**
Diabetes	380	7.7	86	19.0	2.81	1.71–3.35	**<.001**	2.29	1.66–3.17	**<.001**

p values < 0.05 are noted in bold

AYA adolescent and young adult, *OR* odd ratio, *Wt%* weighted percentage

*Adjusted for marital status, education, income, and insurance

^18 controls are missing due to missing data on chronic conditions

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Table 3

Model-adjusted odds of having a chronic condition by sociodemographic factors among Black AYA cancer survivors and age- and sex-matched controls

		Black Controls					Black AYA cancer survivors				
		Having a CHC(s)		95% CI		*p*	Having a CHC(s)		95% CI		*p*
		Wt%	aOR	Lower	Upper		Wt%	aOR	Lower	Upper	
Age at survey	18–39 years	19.0	Ref				57.0	Ref			
	40–64 years	34.7	2.53	1.83	2.78	**<.001**	82.8	3.76	2.04	6.94	**<.001**
	65–85 years	52.7	3.43	2.60	4.52	**<.001**	97.1	23.11	4.86	109.96	**<.001**
Sex	Female	33.6	1.13	0.91	1.40	0.280	76.6	0.64	0.29	1.44	0.280
	Male	25.3	Ref				77.8	Ref			
Marital status	Married or living with partner	20.6	Ref				71.0	Ref			
	Divorced/separated/widowed	52.0	2.71	2.24	3.28	**<.001**	86.1	1.55	0.74	3.23	0.410
	Never married	31.7	1.90	1.55	2.34	**<.001**	71.3	1.23	0.61	2.50	0.780
Education	Less than bachelor's degree	33.2	Ref				78.3	Ref			
	Bachelor's degree or higher	28.0	1.11	0.90	1.37	0.32	70.3	0.53	0.24	1.18	0.120
Family income	Less than $25K	43.8	1.90	1.55	2.32	**<.001**	82.4	1.09	0.54	2.24	0.950
	$25K-$44,999	31.9	1.61	1.27	2.05	**<.001**	67.7	0.68	0.25	1.94	0.660
	$45K or higher	22.3	Ref				74.0	Ref			
Insurance	Uninsured	22.8	0.65	0.51	0.84	0.002	68.6	1.20	0.52	2.78	0.860
	Public	44.5	1.23	1.01	1.50	0.080	85.0	1.84	0.90	3.77	0.170
	Private	27.7	Ref				71.1	Ref			

p values < 0.05 are noted in bold

aOR adjusted odds ratio, *AYA* adolescent and young adult, *CI* confidence interval, *CHC* chronic health condition, *ref* reference, *Wt%* weighted percentage

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Table 4

Model-adjusted odds of chronic health conditions by sociodemographic factors in Black AYA cancer survivors compared with age- and sex-matched controls

			95% CI		*p*
		OR	**Lower**	**Upper**	
Age at survey	18–39 years	5.06	3.13	8.17	**<.001**
	40–64 years	8.45	5.60	12.75	**<.001**
	65–85 years	34.07	7.54	153.92	**<.001**
Sex	Male	14.57	6.47	32.79	**<.001**
	Female	8.30	5.66	12.18	**<.001**
Marital status	Married or living with partner	12.22	6.96	21.46	**<.001**
	Divorced/separated/widowed	6.98	4.03	12.1	**<.001**
	Never married	7.90	4.42	14.09	**<.001**
Education	Less than bachelor's degree	10.70	7.10	16.14	**<.001**
	Bachelor's degree or higher	5.11	2.45	10.68	**<.001**
Family income	Less than $25K	7.21	4.47	11.62	**<.001**
	$25K-$44,999	5.31	2.27	12.42	**<.001**
	$45K or higher	12.51	7.09	22.09	**<.001**
Insurance	Uninsured	13.41	6.07	29.63	**<.001**
	Public	10.87	6.07	19.47	**<.001**
	Private	7.27	4.53	11.66	**<.001**

p values < 0.05 are noted in bold

OR odds ratio, *AYA* adolescent and young adult, *CI* confidence interval